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NEU DAMG7350 (Spring 2026)

ASSIGNMENT: Module 3 – Lab: Implement BitLocker

Introduction

Completing this BitLocker lab required significantly more setup than the instructions suggested, especially for students using Apple Silicon systems. I initially attempted to perform the lab using UTM virtualization on macOS, but Windows 11 ARM64 presented immediate roadblocks. I couldn't obtain a stable ISO from Microsoft or UUP Dump—multiple builds failed to install cleanly, and the images lacked the reliability needed for enterprise security labs. After extensive troubleshooting and testing, I ended up purchasing and configuring a dedicated Windows 11 laptop. Only after that did the TA share a workaround using the Windows Insider Preview ARM64 build, which finally allowed the VM to boot reliably.

Even once Windows ARM64 was running, it became clear that relying on virtualization would introduce unpredictable friction throughout the course. Some enterprise features behaved inconsistently, and later labs required Windows Server components that couldn't run on ARM at all. I decided to maintain two systems for the class: a physical Windows 11 Pro/Enterprise laptop for guaranteed compatibility, and the ARM VM as a secondary environment for experimentation. This setup ensures reliability, but it also highlights a broader issue: the course should explicitly require access to a Windows 11 Pro or Enterprise system before the term begins. Apple Silicon virtualization is not a dependable substitute, and students without a native Windows environment are at a disadvantage.

Procedure

1. Confirm System Readiness: Verified that the system met the requirements for BitLocker. File Explorer showed available drives (C:, E:, D:, F:), with sufficient free space.

2. Launch BitLocker Setup: Opened Control Panel → System and Security → BitLocker Drive Encryption. The interface showed C: waiting for activation and E: in the process of encrypting.

3. Begin Encryption on E: Drive: Selected password unlock method and initiated encryption. Entered and confirmed password.

4. Monitor Encryption Progress: Observed encryption status and available management options, including recovery key backup and password management.

5. Verify Encryption Completion: Used Disk Management to confirm E: was encrypted and marked as Healthy.

6. Confirm Final Drive Status

Validated drive status in File Explorer. E: displayed a padlock icon, indicating full encryption.

Results

The BitLocker configuration completed successfully. The E: drive was encrypted using password-based authentication, and the recovery key was backed up. Encryption status was confirmed through both the BitLocker control panel and Disk Management. The padlock icon in File Explorer confirmed full protection.

The system environment was also stabilized for future labs. OneDrive folder redirection was disabled, local paths were restored, and unnecessary software was removed. These changes ensure that future labs can be completed without interference from domain policies or default system behaviors.

Discussion

The main challenges in this lab stemmed from environmental setup, not BitLocker itself. Apple Silicon virtualization introduced compatibility gaps that consumed time and created uncertainty.

The course materials did not clearly communicate the limitations of ARM-based Windows environments, leading to avoidable delays.

Windows 11 defaults—especially OneDrive integration—created additional obstacles.

Redirected folders and cloud-first save defaults interfered with the controlled environment needed for security labs. Addressing these issues early was essential for accurate results and reproducibility.

Despite these challenges, the lab demonstrated the importance of understanding both technical and environmental factors in enterprise security configurations. BitLocker performed as expected once the environment was properly prepared.

Conclusion

This lab reinforced the importance of working within a stable, supported Windows environment when performing enterprise security tasks. While virtualization on Apple Silicon can be useful for experimentation, it is not a reliable substitute for a native Windows system when working with features like BitLocker or Windows Server components. After transitioning to a dedicated Windows laptop and resolving OneDrive-related issues, the BitLocker configuration proceeded smoothly and produced verifiable, compliant results.

Documenting environmental challenges and workarounds proved essential, as these factors significantly impact the success of security labs. With the environment now stabilized, future labs should proceed more efficiently and with fewer unexpected obstacles.

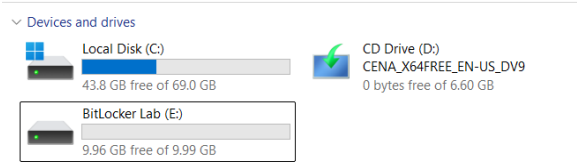
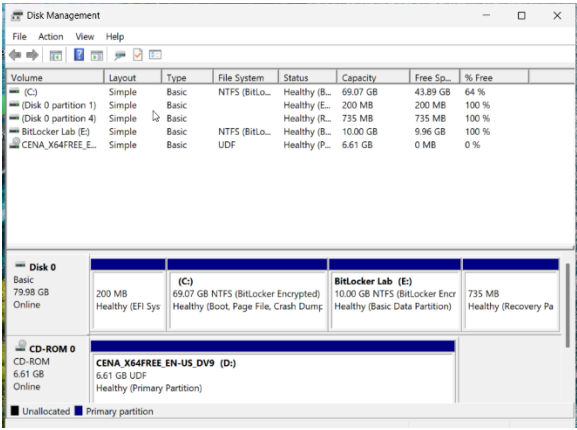
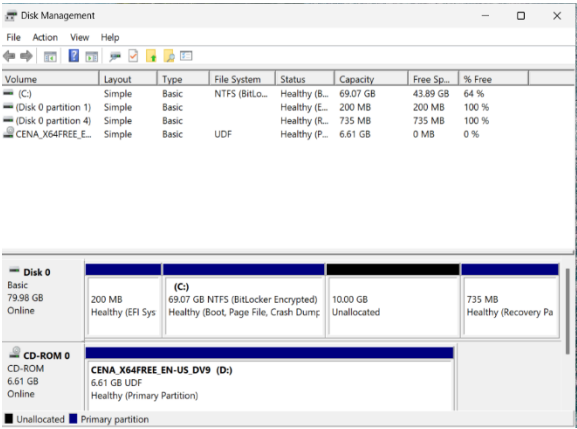
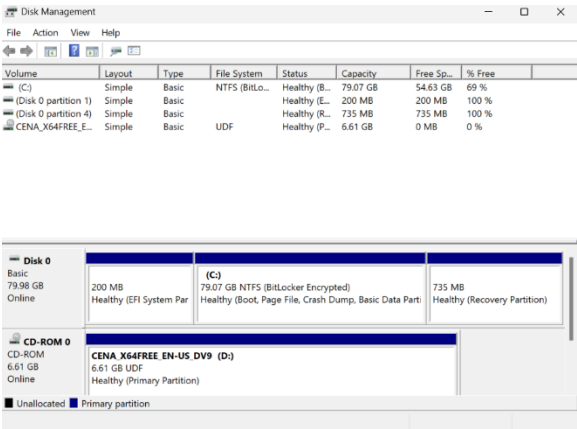
Methodology

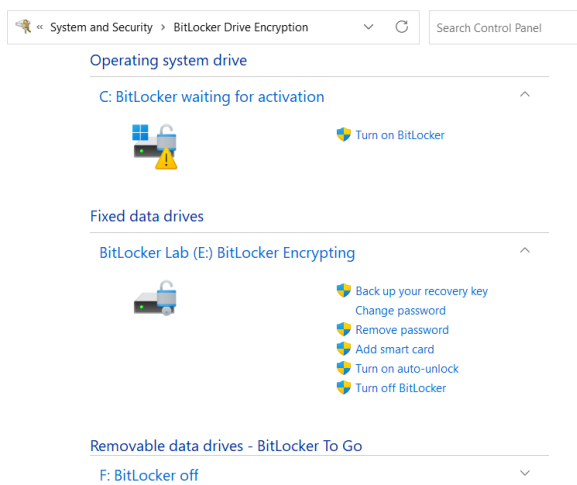
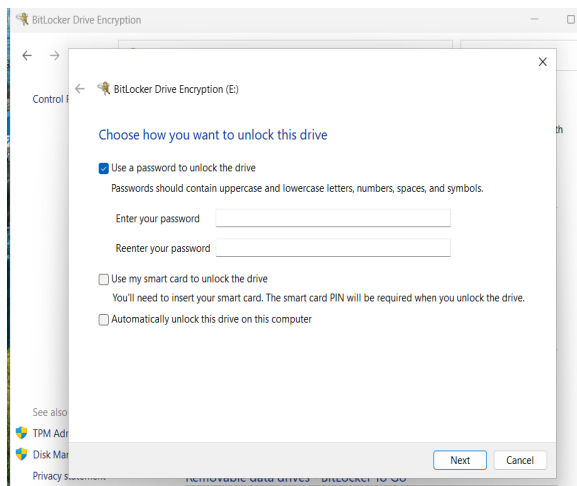
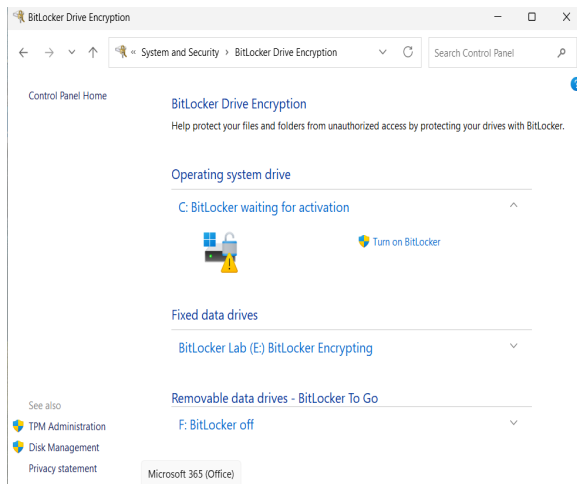
This lab was completed using a combination of hands-on system configuration, iterative troubleshooting, and AI-assisted research. After confirming that virtualization on Apple Silicon was not sufficient, I transitioned to a dedicated Windows 11 Pro laptop. During setup, I encountered issues with system defaults, OneDrive folder redirection, and preinstalled software. I used Copilot to research solutions, validate configuration steps, and generate PowerShell commands to remove unwanted applications and restore local folder paths. Copilot also helped interpret ambiguous instructions, confirm expected system behavior (such as encryption status), and ensure that each step aligned with enterprise best practices. The TA's guidance on ARM64 installation was incorporated into the methodology but is not included in the reference list, per APA guidelines.

References

- Sawant, A. (2025). *3_Lab - Bitlocker.docx* [Unpublished course document]. Northeastern University.
- Sawant, A. (2025). *Module 3: BitLocker Lab* [Course module]. Canvas@Northeastern. <https://northeastern.instructure.com/>

APPENDIX: Screenshots of BitLocker Install Process





▼ Devices and drives



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